

# RadiX Dose

v2.2

*A reference absorbed-dose calculator for external beam radiotherapy.*

- **Reference Dosimetry** — release-ready, verified against several known cases (use with caution)
- **TPS Commissioning** — in development, preview only

RadiX Dose turns a chamber reading into a verified absorbed dose to water — and keeps every calculation, with the equipment and calibrations behind it, in one auditable place. It is built for the medical physicist doing reference dosimetry and periodic output checks at the linac.

It runs natively on iOS (iPhone and iPad) and macOS, works fully offline, and stores everything on-device. Under the hood it is one shared equipment library — machines, electrometers, detectors and beams (photon and electron) — and a single dose engine that computes the reference absorbed dose to water, presented as **two features**, of which only the first is clinically ready.

## 01 Two features, one engine

You pick a feature from the home menu (*Choose a Feature*). Both draw on the same facilities, machines, detectors and protocols — all defined by you in the **Settings** menu — but they sit at very different stages of maturity, and the app colours them accordingly: **blue** for the released tool, **gold** for the one still in development.

### RELEASE-READY

#### Reference Dosimetry

The core of the app. Compute absorbed dose to water from a chamber reading under IAEA TRS-398 or AAPM TG-51, check output against expected, and save an auditable record. Fully implemented and verified against several known reference cases — always confirm with your own independent check.

### IN-DEVELOPMENT

#### TPS Commissioning

Point-dose validation of a treatment planning system against measurement, per IAEA TRS-430 — using the reference dose from IAEA TRS-398 or AAPM TG-51. **A working preview, not a validated clinical tool** — the workflow and acceptance logic are still settling. Do not rely on it for release decisions.

## 02 The setup model — build your clinic once

Everything a calculation needs is described once, as a nested library under **Settings**. A result is only as trustworthy as the equipment record behind it, so this hierarchy is the app's backbone. One facility and one machine are *active* at a time; the whole app scopes to them.

### SETUP HIERARCHY — SETTINGS

#### **FACILITY** Hospital · department

#### **MACHINE** Linac · code of practice · setup & dose-reporting point

##### Detector library

- Chamber from a built-in catalog (41 models, 6 manufacturers) or user-defined
- Manufacturer, serial number, cavity volume & length
- $N_{D,w}$  certificate + date + validity
- $^{60}\text{Co}$  or cross-calibration route — offered for every chamber
- Correction factors stored per chamber & energy

##### Electrometer library

- Manufacturer, model, serial number
- Optional calibration factor  $k_{elec}$  (default 1.000)
- Pairs with the chamber's  $N_{D,w}$  certificate

##### Beam library

- Photon & electron energies (incl. FFF)
- Beam-quality index per protocol ( $TPR_{20,10} \cdot \%dd(10)_x \cdot R_{50}$ )
- **Calibration conditions:** field size, delivered MU & expected dose
- PDD / TMR at the reference depth (for  $d_{max}$  normalisation)

Each machine carries its own **code of practice**, **SSD/SAD setup**, and **dose-reporting point** (see section 03). Photon-only machines (Halcyon, Unity) and special-field machines (CyberKnife, TomoTherapy) are recognised and route their dosimetry accordingly — though the special-field machines are not yet fully validated, so use them with care; your feedback is welcome and will be applied where needed. Every delete — facility, machine, detector, beam — is confirmed before it cascades.

### Calibration expiry, watched for you

Each certificate carries a date and a validity period. Detector cards turn **yellow at 3 months**, **red at 1 month**, and read **EXPIRED** past due — a warning only, never a block. You decide what to calculate with.

### 03 The calculation pipeline

This is the engine. A raw chamber reading is corrected for every influence quantity, scaled by the chamber's calibration and beam-quality factor, and turned into absorbed dose to water — then compared to what the beam *should* deliver.

#### FROM READING TO DOSE

##### INPUT

###### Reading + conditions

Up to 3 charge readings (their mean,  $M$ ), temperature & pressure, chamber & beam.



##### CORRECT

###### Influence factors

 $K_{TP}$  $K_{Pol}$  $K_S$  $K_{Elec}$  $K_{P/W}$  $K_{Vol}$ 

##### SCALE

###### Calibration & quality

 $N_{D,w}$  $K_Q$ 

Chamber's dose-to-water factor and its beam-quality correction.



##### RESULT

###### Absorbed dose $D_{w,Q}$

Dose to water, reported at the machine's chosen point (section below).



##### CHECK

###### Output vs expected

Measured cGy/MU against the beam's calibration condition → % deviation.

corrected reading

$$M = M_{raw} \cdot K_{TP} \cdot K_{Pol} \cdot K_S \cdot K_{Elec}$$

absorbed dose to water

$$D_{w,Q} = M \cdot N_{D,w} \cdot K_Q \cdot K_{P/W} \cdot K_{Vol}$$

Every factor is resolved automatically from your libraries and the active protocol, or entered on the **Factors** tab. Divisions are guarded, so a missing input *withholds* the dose rather than producing a

wrong one.

### Where the dose is reported — your choice, per machine

Set each machine's **dose-reporting point** alongside its SSD/SAD setup: report at  $d_{\max} / z_{\text{ref}}$  — scaling the reference-depth dose by **PDD** (SSD) or **TMR** (SAD) — or **at the reference (calibration) depth** itself. The chosen depth travels with every saved record.

## How the deviation reads

Output is compared as a rate — measured **cGy/MU** against the beam's calibration condition, so a per-MU calibration and a 100-MU calibration compare correctly. The % difference sits on a bell-curve gauge with a tolerance band:

■  $\leq 2\%$  — within tolerance    ■  $\leq 3\%$  — investigate    ■  $> 3\%$  — out of tolerance

### Small & special fields

For machines with no 10×10 reference field (CyberKnife, TomoTherapy), the app routes to **TRS-483** small-field dosimetry ( $k_{Q_{\text{MSR}}, \text{fref}}$ ). Off-reference geometries withhold the dose until the reference field is restored. Please use these special-field machines with care — they are not yet fully validated, and your feedback is welcome and will be applied where needed.

## 04 The correction factors

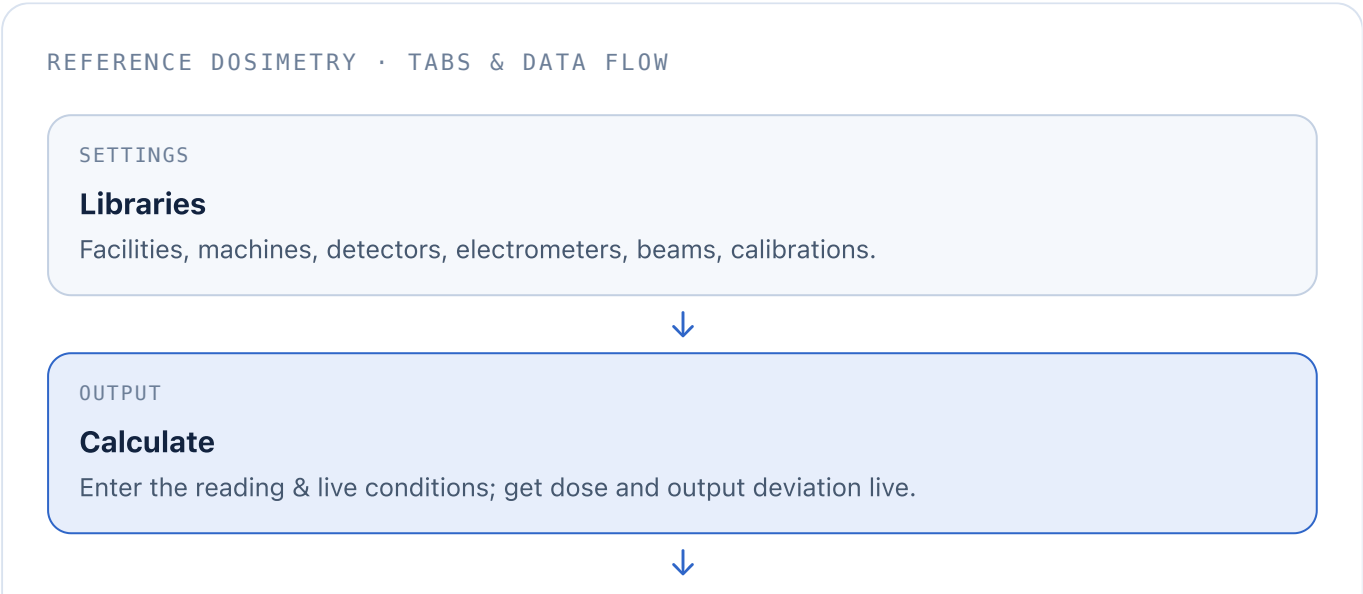
Nine quantities can enter a calculation. The app resolves each where it can, and the **Factors** tab is where you measure and *lock* the rest — stored per chamber and energy, so the same probe carries its own factors across every beam. Symbols follow each protocol's own notation (IAEA **K** / AAPM TG-51 **P**).

| SYMBOL                | CORRECTS FOR                       | WHERE IT COMES FROM                                           |
|-----------------------|------------------------------------|---------------------------------------------------------------|
| $K_{TP} / P_{TP}$     | Air density                        | Live from temperature & pressure on Output                    |
| $K_{PoL} / P_{PoL}$   | Polarity effect                    | Factors — opposite-polarity readings, per chamber+energy      |
| $K_S / P_{Ion}$       | Ion recombination                  | Factors — two-voltage method                                  |
| $K_{Elec} / P_{Elec}$ | Electrometer calibration           | Electrometer library (default 1.000)                          |
| $K_{P/W}$             | Waterproof sleeve / phantom        | Factors — per chamber+energy                                  |
| $K_{Vol}$             | Volume averaging (FFF beams)       | Automatic — needs cavity length (TRS-398 Rev.1)               |
| $N_{D,w}$             | Absolute dose-to-water calibration | Detector library ( $^{60}\text{Co}$ or cross-cal certificate) |
| $K_Q$                 | Beam quality vs $^{60}\text{Co}$   | Protocol tables, or a custom / overridden value               |
| $P_{Gr}$              | Gradient (electron)                | Factors — AAPM TG-51 electron only                            |

Factors are **locked to apply**: a measured factor stays out of the dose calculation until you lock it, so a half-entered value never silently changes a result. The Factors tab is a set of sheets on iPhone and a single-open accordion on iPad and Mac.

## 05 Module map — where data flows

The Reference Dosimetry feature is four tabs. Data flows one way: configure once, calculate, save, report.



#### RESULTS

##### Save & trend

Auditable records, date filters, a scrolling trend chart.



#### REPORT · BACKUP

##### Share & preserve

A4 PDF report; export/import everything as one file.

**Factors** sits alongside Output, feeding the corrections of section 04 into the calculation. When a chamber or beam changes on Output, the charge readings clear automatically so a stale number can't carry over.

## A measurement, start to finish

### 1 Configure once

In Settings, add your facility, machine (protocol, setup, dose-reporting point), chambers with their  $N_{D,w}$  certificates, and beam energies with their calibration conditions.

### 2 Read & enter

On Output, pick the chamber and beam, then enter temperature and pressure (from live telemetry) and up to three charge readings.

### 3 Lock & read

Absorbed dose and % deviation appear live on the gauge; lock any outstanding factors on the Factors tab.

### 4 Save & report

Save the record, then filter, chart, and export a PDF — or back the whole database up to one file.

## 06 Protocols & beam quality

Each machine is assigned a code of practice; the engine routes the beam-quality factor and reference conditions accordingly. Three are live, and each expects a different beam-quality index:

| PROTOCOL            | PHOTON QUALITY | ELECTRON | NOTES                              |
|---------------------|----------------|----------|------------------------------------|
| IAEA TRS-398 Rev.1  | $TPR_{20,10}$  | $R_{50}$ | Default · auto $K_{vol}$ for FFF   |
| IAEA TRS-398 (2006) | $TPR_{20,10}$  | $R_{50}$ | Tabulated $K_Q$ · 5 or 10 cm depth |
| AAPM TG-51          | $\%dd(10)_x$   | $R_{50}$ | Muir & Rogers 2014 electron $K_Q$  |

- **Switch safely.** Change a machine's protocol and any beam missing the new index (e.g. a  $\%dd(10)_x$  where only  $TPR_{20,10}$  exists) is flagged red with a clear message, before it can produce a wrong  $K_Q$ .
- **Custom  $K_Q$ .** Where a protocol's own tables can't answer — an unlisted chamber, or a beam quality outside the fitted range — supply your own value, stored per protocol and energy, never mixed between them.
- **Override  $K_Q$ .** When the tables *do* answer but you have an independently derived value you'd rather use, override the auto value from the  $K_Q$  card. Overriding is discouraged and clearly flagged — the dose is tagged "**Overridden**" so its provenance is never hidden. Custom (a gap-filler) and Override (a replacement) stay distinct.
- **Coming soon.** DIN 6800-2 and NCS Report 18 appear in the picker, disabled for now.

## 07 Managing the results

A calculation isn't finished until it's recorded. The management side is as much the point as the math — every saved result keeps the full factor snapshot that produced it, down to the charge reading.

- **Records** — every saved calculation with its complete factor set, the mean charge reading  $M$ ,  $N_{D,w}$ , and conditions; filterable by date (presets or a custom calendar range) and machine.
- **Trend chart** — a scrolling day-by-day view of output deviation with per-day labels, so drift is visible at a glance.
- **Report** — an A4 landscape PDF (trend chart + summary table, incl.  $M$  and  $N_{D,w}$ ) built from the records you select.
- **Backup** — export everything (facilities, machines, detectors with certificates, beams with factors, saved results) to one file, and import it on another device; the format tolerates version changes.

## 08 TPS Commissioning IN-DEVELOPMENT

### ⚠ Preview feature — not for clinical release

This feature is actively being built. Its workflow, tolerances, and acceptance logic are **not yet validated for clinical commissioning decisions**. Use it to explore the concept — not to sign off a treatment planning system. Reference Dosimetry is the released tool, verified against several known cases; this is not.

The idea: reuse the very same dose engine to turn a chamber reading at a point into a measured dose  $D_{\text{meas}}$ , compare it to the TPS-predicted dose, and evaluate the deviation against a tolerance — the standard point-dose validation of IAEA TRS-430, using the reference dose from IAEA TRS-398 or AAPM TG-51.

#### COMMISSIONING FLOW – PREVIEW

##### ORGANISE

###### Test Group

A TPS + algorithm + version, per machine.



##### DEFINE

###### Test

A test type, energy, field size & setup, with measurement points.



##### MEASURE

###### Point → $D_{\text{meas}}$

Chamber readings run through the reference dose engine.



##### EVALUATE

###### vs TPS · pass/fail

Deviation against tolerance; confidence limit across a group.

## Four test types

- **Central-axis (CAX) point dose** — the on-axis test.
- **Off-axis point dose (in field)** — a test inside the field edges.
- **Out-of-field point dose** — a test outside the beam.
- **Output factors** — the field-size output check.

## How a point is judged

Each point takes up to three chamber readings (mean →  $D_{\text{meas}}$ ) and the TPS-predicted dose. The deviation is plain for every point, evaluated against a tolerance you pick (1, 1.5, 2, 3 or 4 %), and a confidence limit summarises a whole group:



deviation

$$\delta = 100 \cdot (D_{\text{TPS}} - D_{\text{meas}}) / D_{\text{meas}}$$

confidence limit

$$\text{CL} = |\text{mean } \delta| + 1.5 \cdot \text{SD}$$

Because  $D_{\text{meas}}$  comes from the same verified chain as Reference Dosimetry, the measured side is sound — it is the surrounding commissioning workflow and acceptance logic that is still in development.

## 09 Your responsibility

RadiX Dose is a calculation aid. It launches behind a User Agreement you accept before clinical use, and the calculating physicist remains responsible for verifying every result against an independent method. The app warns — expired calibrations, missing inputs, off-reference geometry, mismatched beam quality, an overridden  $K_Q$  — but never overrides your judgement, and never blocks a calculation you choose to make.

### In one line

**Reference Dosimetry** is ready for clinical use — to check and track output — with normal physics verification. **TPS Commissioning** is a preview of where RadiX Dose is heading — explore it, don't rely on it yet.

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RadiX Dose v2.2 · User Guide. A brief companion to the app — pipeline diagrams, factor and protocol reference, feature overview. Developer & licence holder: Oğuzhan Ayrancıoğlu, Medical Physicist.